



SmartInvest SDG — Profit-Optimized, SDG-Aligned Portfolios

An intelligent cross-market platform that combines financial performance, ESG indicators and UN SDG alignment to recommend portfolios that balance returns with measurable sustainability impact across India and global markets.



Problem — Why Current Investing Falls Short

Returns-Only Focus

Most investors optimise for short-term gains and overlook environmental and social harm created by portfolio companies.

Fragmented Markets

Indian and global markets are siloed — no unified view combining performance and sustainability for cross-market decision making.

Scattered ESG Data

ESG/SDG information is inconsistent, incomplete and hard for retail investors to interpret.

Profit vs Impact Trade-offs

Investors lack tools to quantify the trade-off between profitability and long-term sustainable impact.



Our Solution — One Dashboard, Dual Goals

SmartInvest SDG integrates financial metrics, multi-source ESG data and SDG alignment into a single cross-market dashboard. It generates personalised, profit-optimized portfolios that minimise negative environmental and social impact while remaining market competitive.



Intelligent Algorithms

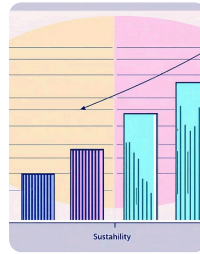
- Multi-objective optimisation (profit + ESG penalties)
- Pareto frontier analysis to show efficient trade-offs
- GRG and convex solvers for constrained optimisation
- Proxy estimation where ESG data is missing

Investor Features — Clear, Actionable Insights



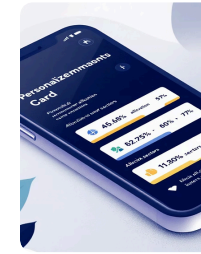
Stock-level SDG Insights

Explainable mappings of company activities to specific SDG targets, with confidence scores and evidence links.



Trade-off Visualiser

Interactive slider to prioritise profit or sustainability and instantly see portfolio shifts and impact metrics.



Personalised Recommendations

Candidate portfolios tailored to risk preference, impact goals, and market exposure — with allocation rationale (example: 40% clean energy, 30% healthcare, 30% tech).

Evidence Base — Zhang (2025) Explained

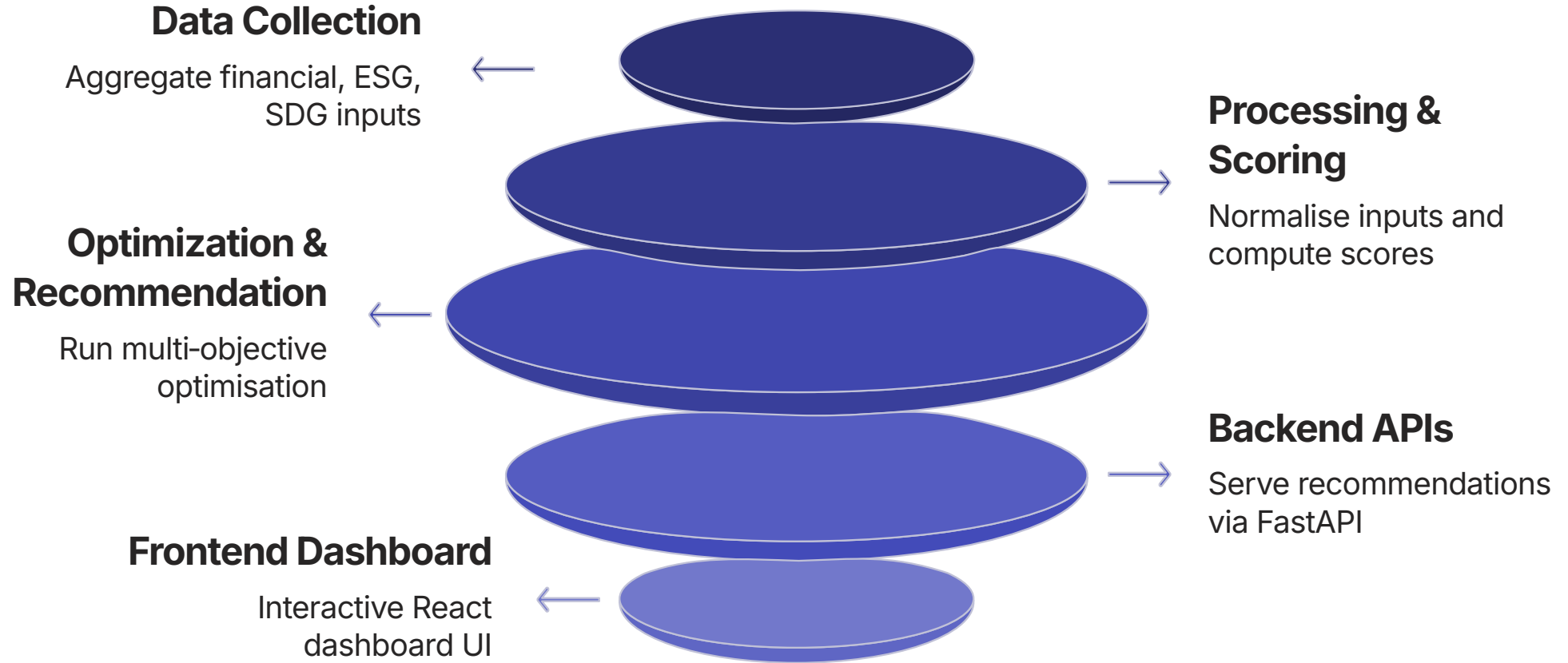
Zhang (2025) empirically shows ESG-constrained portfolios can outperform market indices on a risk-adjusted basis while offering smoother returns. Key takeaway: ESG constraints reduce volatility with modest trade-offs in peak returns — directly supporting SmartInvest's multi-objective approach.

Why it matters

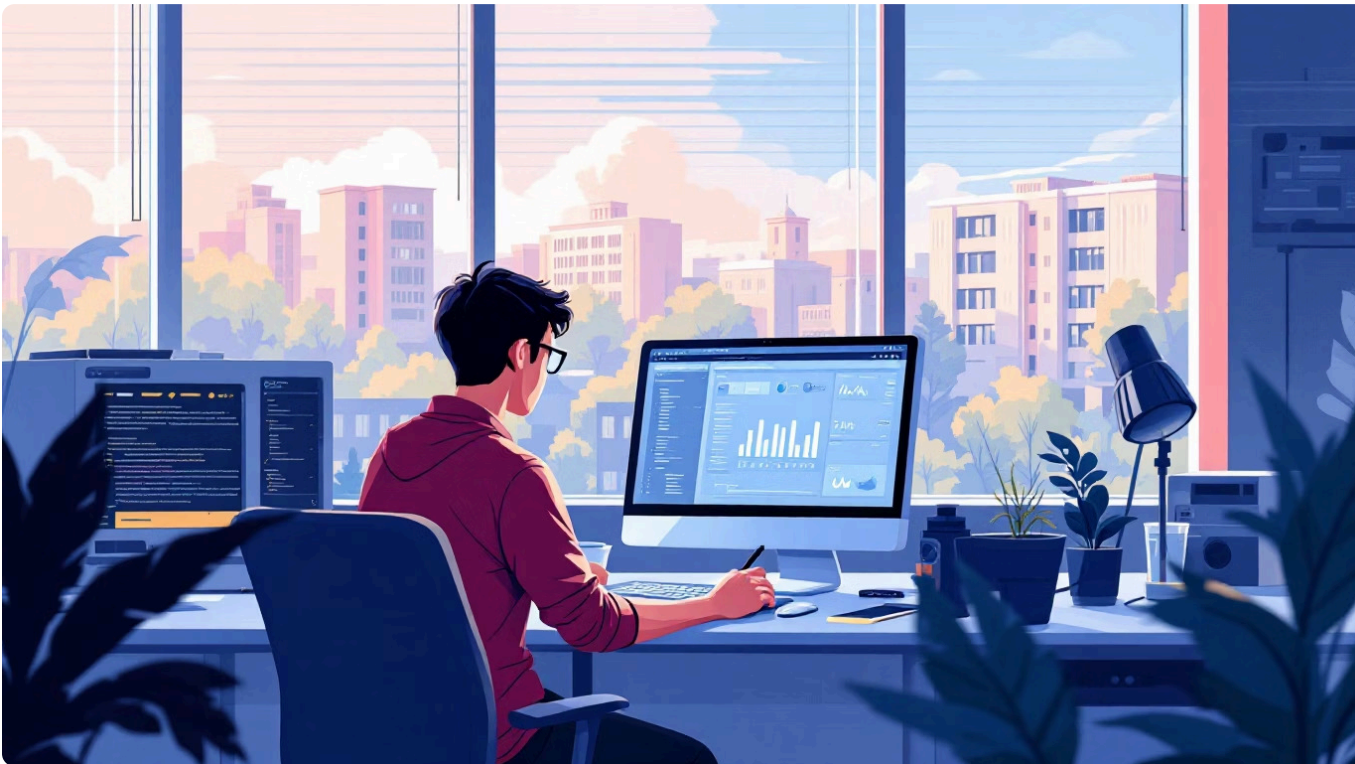
Validates optimization approach (GRG) and highlights data limitations our platform addresses.

Limitations

NASDAQ-only sample, limited universe, single ESG score — gaps SmartInvest remedies via multi-market coverage and granular SDG mapping.



The system unifies financial, ESG and SDG inputs → normalises and scores them → runs multi-objective optimisation → serves recommendations via FastAPI to an interactive React dashboard.



Tech Stack (Practical & Scalable)



Data & Models

Python, Pandas, NumPy, SciPy, CVXPY, Scikit-learn



Backend & DB

FastAPI, PostgreSQL, Redis (cache)



Frontend

React, Chart.js / D3 for visualisers, responsive UI for retail users

Implementation Roadmap — 7 Weeks

01

Weeks 1–2: Data & Schema

Collect Indian + global market data, assemble ESG samples, build DB schema and processing pipelines.

03

Week 4: Optimization Engine

Implement multi-objective solvers, GRG or convex alternatives, generate candidate portfolios and Pareto sets.

02

Week 3: ESG–SDG Scoring

Normalize indicators, create SDG mapping rules and proxy estimators for missing data.

04

Weeks 5–7: API, UI, Testing

Build FastAPI endpoints, React dashboard with trade-off slider, user testing and deploy optionally on cloud.

Next Steps & Ask

Pilot SmartInvest SDG with a curated basket of Indian and international stocks, validate impact metrics against SDG targets, and secure early adopters among retail & institutional investors. Seeking partnerships for ESG data, a pilot user group, and seed technical resources.

